THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET D

Chapters 13 & 14: **Statistics, Probability** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The median of 3, 5, 7, 9, 11 is: 1

(a) 9 **(b)** 5 **(c)** 35 **(d)** 7

2. The class width of the classes 10–20, 20–30, … is: 1

(a) 15 **(b)** 20 **(c)** 10 **(d)** 5

3. If the mode is 30 and the median is 26, the mean is: 1

(a) 32 **(b)** 28 **(c)** 24 **(d)** 22

4. A student states: "The step-deviation method gives a different mean from the direct method." The correct response is: 1

(a) It is always larger **(b)** Incorrect — both give the same mean

(c) Correct **(d)** Cannot be determined

5. For the class mark of 0–20, Student A says 10; Student B says 20. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. The probability of a number greater than 6 on a die is: 1

(a) 1 **(b)** $\frac{5}{6}$ **(c)** $\frac{1}{6}$ **(d)** 0

7. The probability of drawing a non-face card from a deck of 52 cards is: 1

(a) $\frac{10}{13}$ **(b)** $\frac{9}{13}$ **(c)** $\frac{3}{13}$ **(d)** $\frac{1}{2}$

8. Two coins are tossed. The probability of two tails is: 1

(a) 1 **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{4}$

9. A student states: "P(at least one head in two tosses) $= \frac{1}{2}$." The correct response is: 1

(a) It is $\frac{1}{4}$ **(b)** Correct

(c) Incorrect — it is $\frac{3}{4}$ **(d)** Cannot be determined

10. A bag has 5 red and 5 blue balls. Student A says P(red) $= \frac{1}{2}$; Student B says $\frac{1}{5}$. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mean of: 2

Class	0–6	6–12	12–18
Frequency	3	5	2

12. Find the mode of: 2

Class	20–30	30–40	40–50
Frequency	4	10	6

13. A number is chosen from 1 to 10. Find the probability that it is prime. 2

14. A letter is chosen from MISSISSIPPI. Find P(S) and P(I). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the median height: 3

Height (cm)	140–145	145–150	150–155	155–160	160–165
Students	3	7	10	6	4

16. Two dice are thrown. Find P(sum 12), P(sum odd) and P(product 12). 3

17. Find the mean by the step-deviation method: 3

Age (yrs)	25–35	35–45	45–55	55–65	65–75
Persons	6	11	7	4	2

OR

Find the mode of: 3

Class	1–4	4–7	7–10	10–13	13–16	16–19
Frequency	6	30	40	16	4	4

18. A jar has 54 marbles — blue, green and white. P(blue) $= \frac{1}{3}$ and P(green) $= \frac{4}{9}$. Find the number of white marbles. 3

OR

Cards numbered 11 to 60 are in a box. One is drawn. Find P(perfect square) and P(divisible by 5). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the mean and median of the runs scored: 5

Runs	0–20	20–40	40–60	60–80	80–100
Players	7	10	15	12	6

OR

A student claims the mean of the data below is 40. 5

(i) Find the actual mean.
(ii) Comment on the claim.
(iii) Find the median.

Class	0–20	20–40	40–60	60–80	80–100
Frequency	12	18	10	6	4

20. **Case study:** The blood groups of 30 students are: A — 9, B — 6, AB — 3, O — 12. A student is chosen at random. Find 5

(i) P(O)
(ii) P(AB)
(iii) P(not A)
(iv) P(A or B).

OR

Two coins are tossed. Find 5

(i) P(both heads)
(ii) P(exactly one head)
(iii) P(no head).
(iv) Verify that the probabilities add to 1.
(v) A student says ‘one head’ is as likely as ‘two heads’. Comment.

END OF QUESTION PAPER — Total: 40 Marks

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